Akanksha Wagh

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SUMMARY

Data Scientist with 3+ years of experience turning complex business questions into clear, actionable strategies. I've built AI agents, customer segmentation models that powered product launch campaigns, designed experimentation frameworks that shaped website-wide product decisions, and automated executive reporting with AI tools — all while working closely with marketing, product, and operations teams.

EDUCATION

Master of Science in Data Science - Stevens Institute of Technology — August 2023 - May 2025
Bachelor of Technology in Information Technology - Pune University – JSPM College of Engineering — August 2019 - June 2023

WORK EXPERIENCE

Data Scientist – Retail & Marketing, Compass Group USA — February 2024 - May 2025
Tech: *Python, SQL, Power BI, Azure Synapse, Scikit-Learn, NLP*

- Developed **an AI agent** that generated **weekly engagement OKRs** by extracting campaign and CRM data from Microsoft Dynamics 365 via MCP server queries, **automating KPI analysis** and improving reporting structure.
- Built **customer segmentation** and **demand forecasting** models to optimize retail performance — **tracking KPIs** like inventory turnover, sell-through rate, and demand accuracy to improve inventory **decision-making by 25%.**
- Analyzed **3,100+ customer** feedback records using **sentiment analysis**, directly supporting vendor selection and replacement decisions.
- Performed **market basket analysis** and sales velocity modeling to identify high-performing and slow-moving SKUs, informing pricing, bundling, and **procurement strategies** that improved **sell-through efficiency by 30%.**

Data Analyst, Stevens Institute of Technology — May 2024 - August 2024
Tech: *Python, PySpark, LangChain, GPT-4, MLflow*

- Analyzed **IT support ticket data** for Attila, an AI chatbot, to identify **recurring** student **issues** — including campus WiFi and email activation — translating findings into chatbot conversation flows that **reduced human agent dependency by 37%.**
- Designed and tracked **real-time analytics dashboards** monitoring intent accuracy, escalation rate, and response quality — providing stakeholders with **continuous visibility into chatbot performance**.
- Built automated performance monitoring pipelines — **improving resolution accuracy by 20%** and **reducing escalation rate by 40%.**

Business Data Analyst, Tata Motors Limited — January 2023 - June 2023
Tech: *PySpark, Python, PostgreSQL, Django REST, Power BI, ML Analytics*

- Engineered analytics pipelines for Drishti 2.0, **an internal employee portal** integrating data from **30,000+ employees** — delivering personalized **workforce insights** and improving **planning efficiency** across the organization.
- Built Power BI dashboards and performed **statistical analysis** on **800K+ IT support ticket** records **across 5 manufacturing plants**, identifying recurring issues and **automating high-frequency problems** — reducing manual reporting effort by 37% and improving operational planning efficiency by 21%.

Marketing Data Analyst, Nykaa — June 2021 - December 2022
Tech: *Python, SQL, Snowflake, dbt, AWS S3, Power BI*

- Designed and maintained **ETL pipelines** processing **1M+ records** monthly, integrating transactional, inventory, and customer behavior data to support forecasting and reduce stockout and overstock risk.
- Partnered **cross-functionally** with product, marketing, and merchandising teams to translate **100M+ clickstream** data insights into discount and personalization strategies — boosting **campaign ROI by 44%** and improving **conversion rate by 32%.**
- Implemented **A/B testing** framework to evaluate recommendation algorithm improvements, product positioning, and pricing strategies — enabling data-driven merchandising decisions that increased **user engagement by 36%.**

PROJECTS & PUBLICATIONS

Personal Portfolio Website with Embedded AI Agent [Talk to my Agent] (**Tech:** *HTML, CSS, JavaScript, Anthropic Claude API*)
Built a portfolio website with an embedded AI agent —to answer visitor questions about my experience, skills, and projects in real time.

Forecasting Metro-Level Hiring Trends for Indeed Job Postings (**Tech:** *R, ARIMA, SARIMA, ACF/PACF, AIC/BIC)*
Developed forecasting models on Indeed's daily job posting data to predict NYC hiring trends, achieving 18% improvement in forecast accuracy.

Self-Generated Soccer Highlights using I3D-CNN (**Tech:** *GANs, Data Augmentation, PyTorch, OpenCV, CNNs*)
Built DL event detection model on SoccerNet data with **95% accuracy**, improving highlight-generation precision by **25%**.[Published in IEEE]

TECHNICAL SKILLS

Analytics, Cloud & Tools: SQL, Python, Power BI, Tableau, HTML, CSS, JavaScript, Streamlit, Excel, AWS, Azure, GCP, Airflow
Statistics & Methods: A/B Testing, Regression Analysis, Customer Segmentation, Market Basket Analysis, Forecasting
Data Engineering: PySpark, ETL, Snowflake, dbt, AWS S3, Azure Synapse